

SC1001: Introduction to Computational Thinking

Week 4: Repetition (Looping)

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Slides

Slides and updates will be posted at

boydgraber.org/teaching/SC_1001/

Overview

1. Count temperatures above a threshold with a loop
2. Trace a loop with `continue`, `pass`, and `while-else`
3. Repair a sentinel loop and extend it to track more information
4. Write a looping solution for FizzBuzz
5. Print a pyramid pattern with `for` loops

Discussion 1: Days Above 25°C

Assume we are already given a list called `daily_temperatures` with 30 values.

Write Python code that counts how many *days* have temperature strictly above 25.

```
daily_temperatures = [22, 25, 27, 23, 28, 26, 24, 30, 31, 29,  
                      21, 22, 26, 27, 28, 29, 25, 30, 31, 32,  
                      20, 21, 22, 23, 24, 25, 26, 27, 28, 29]
```

Discussion 1: Hardcoded List Solution

```
daily_temperatures = [22, 25, 27, 23, 28, 26, 24, 30, 31, 29,  
                       21, 22, 26, 27, 28, 29, 25, 30, 31, 32,  
                       20, 21, 22, 23, 24, 25, 26, 27, 28, 29]  
  
days_above_25 = 0  
  
for day in daily_temperatures:  
    if day > 25:  
        days_above_25 += 1  
  
print("Number of days above 25:", days_above_25)
```

Discussion 1: Add-On

Suppose we instead store the temperatures by week:

```
temperature_by_week = {  
    "week 1": [22, 25, 27, 23, 28, 26, 24],  
    "week 2": [30, 31, 29, 21, 22, 26, 27],  
    "week 3": [28, 29, 25, 30, 31, 32, 20],  
    "week 4": [21, 22, 23, 24, 25, 26, 27],  
    "week 5": [28, 29]  
}
```

How would that change the solution?

Discussion 1: About the Dictionary Add-On

```
days_above_25 = 0

for week in temperature_by_week:
    for day in temperature_by_week[week]:
        if day > 25:
            days_above_25 += 1
```

The basic idea stays the same, but now we need a *nested loop*: first over weeks, then over the temperatures inside each week.

Discussion 2: What Is the Output?

```
value = 6
if value % 2 == 0:
    print("first", value)
elif value % 3 == 0:
    print("second", value)
while value <= 9:
    value = value + 1
    if value == 8:
        continue
    else:
        pass
    print("third", value)
else:
    print("fourth", value)
print("fifth", value)
```

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

first 6

Since $6 \% 2 == 0$, the first printed line is first 6.

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

first 6

Value is divisible by three, but it was also divisible by two, so the `elif` is not triggered.

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                     #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

first 6

The loop increments to 7

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)               #L14
print("fifth", value)                    #L15
```

Output

first 6

pass says “do nothing in this block”

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

first 6
third 7

third 7 is printed

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                     #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                    #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

first 6
third 7
skip 8

Increment value to eight

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

first 6
third 7

When `value == 8`, `continue` skips the rest of that iteration.

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)               #L14
print("fifth", value)                    #L15
```

Output

```
first 6
third 7
third 9
```

We increment value to 9, and pass again does nothing, but we print third 9

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)               #L14
print("fifth", value)                    #L15
```

Output

```
first 6
third 7
third 9
third 10
```

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

```
first 6
third 7
third 9
third 10
fourth 10
```

The while-else runs because the loop ends normally when value becomes 10.

Discussion 2: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        pass                             #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

```
first 6
third 7
third 9
third 10
fourth 10
fifth 10
```

The final print happens after the loop, so we also see fifth 10.

Discussion 2: What Is the Output?

```
value = 6
if value % 2 == 0:
    print("first", value)
elif value % 3 == 0:
    print("second", value)
while value <= 9:
    value = value + 1
    if value == 8:
        continue
    else:
        break
    print("third", value)
else:
    print("fourth", value)
print("fifth", value)
```

Discussion 2B: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        break                            #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)               #L14
print("fifth", value)                    #L15
```

Output

first 6

Since $6 \% 2 == 0$, the first printed line is first 6.

Discussion 2B: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                     #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        break                            #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)               #L14
print("fifth", value)                    #L15
```

Output

first 6

The loop increments to 7

Discussion 2B: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        break                            #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)                #L14
print("fifth", value)                     #L15
```

Output

first 6

break goes to end of loop

Discussion 2B: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)                #L5
while value <= 9:                         #L6
    value = value + 1                     #L7
    if value == 8:                        #L8
        continue                         #L9
    else:                                 #L10
        break                             #L11
    print("third", value)                  #L12
else:                                     #L13
    print("fourth", value)                 #L14
print("fifth", value)                     #L15
```

Output

first 6

Do not pass go, do not evaluate else (only if while evaluates to false)

Discussion 2B: Walk Through the Control Flow

```
value = 6                                #L1
if value % 2 == 0:                        #L2
    print("first", value)                 #L3
elif value % 3 == 0:                     #L4
    print("second", value)               #L5
while value <= 9:                         #L6
    value = value + 1                    #L7
    if value == 8:                       #L8
        continue                        #L9
    else:                                #L10
        break                            #L11
    print("third", value)                 #L12
else:                                     #L13
    print("fourth", value)               #L14
print("fifth", value)                     #L15
```

Output

first 6
fifth 7

The final print happens after the loop, so we also see fifth 7.

Discussion 3A: Find and Fix the Errors

The program should:

- count how many input strings contain the letter a,
- stop when the user enters ####,
- and print the total at the end.

```
while True:
    str = input("enter a string: ")
    for letter in str:
        if letter == 'a':
            break
    count += 1
print(count, "strings with letter 'a'")
```

Discussion 3A: Highlight the Errors

```
while True:                                     #L1
    str = input("enter a string: ")             #L2
    for letter in str:                           #L3
        if letter == 'a':                       #L4
            break                               #L5
        count += 1                              #L6
    print(count, "strings with letter 'a'")     #L7
```

count never initialized.

Discussion 3A: Highlight the Errors

```
while True:                                #L1
    str = input("enter a string: ")         #L2
    for letter in str:                      #L3
        if letter == 'a':                  #L4
            break                           #L5
        count += 1                          #L6
    print(count, "strings with letter 'a'") #L7
```

Avoid naming a variable `str`; it shadows a built-in function.

Discussion 3A: Highlight the Errors

```
while True:                                     #L1
    str = input("enter a string: ")             #L2
    for letter in str:                           #L3
        if letter == 'a':                       #L4
            break                                #L5
        count += 1                              #L6
    print(count, "strings with letter 'a'")     #L7
```

There is no sentinel test, so the loop never stops on ####.

Discussion 3A: Highlight the Errors

```
while True:                                #L1
    str = input("enter a string: ")         #L2
    for letter in str:                      #L3
        if letter == 'a':                  #L4
            break                           #L5
    count += 1                              #L6
    print(count, "strings with letter 'a'") #L7
```

This counts characters before a, not strings containing a.

Discussion 3A: Highlight the Errors

```
while True:                                #L1
    str = input("enter a string: ")         #L2
    for letter in str:                      #L3
        if letter == 'a':                  #L4
            break                           #L5
        count += 1                         #L6
    print(count, "strings with letter 'a'") #L7
```

The print should summarize after the loop, not after every input.

Discussion 3A: Fix the Errors

```
count_with_a = 0

text = input("enter a string (enter #### to stop): ")
while text != "####":
    for letter in text:
        if letter == "a":
            count_with_a += 1
            break
    text = input("enter a string (enter #### to stop): ")
print(count_with_a, "strings with letter 'a'")
```

Discussion 3B: Membership Version

Can we solve the repaired problem using direct string membership instead of an inner loop over letters?

Discussion 3B: List Membership

```
count_with_a = 0

text = ''
while text != "####":
    count_with_a += int("a" in text)
    text = input("enter a string (enter #### to stop): ")

print(count_with_a, "strings with letter 'a'")
```

Discussion 3C: List-Comprehension Version

Can we use a list comprehension to print the total number of a's entered across all input strings?

Discussion 3C: List Comprehension

```
text = ''
total_a = 0

while text != '####':
    total_a += sum(1 for x in text if x=='a')
    text = input("enter a string (enter #### to stop): ")

print(total_a, "total letter 'a' characters")
```

Could also do `text.count("a")`

Discussion 4: Write the Python Program

Write a looping solution for FizzBuzz from 1 to 20.

- print FizzBuzz for multiples of both 3 and 5
- otherwise print Fizz for multiples of 3
- otherwise print Buzz for multiples of 5
- otherwise print the number

Discussion 4: One Python Solution

```
for ii in range(1, 21):  
    if ii % 15 == 0:  
        print("FizzBuzz")  
    elif ii % 3 == 0:  
        print("Fizz")  
    elif ii % 5 == 0:  
        print("Buzz")  
    else:  
        print(ii)
```

Can we do it with two if statements?

```
for num in range(1, 20):  
    if num % 3 == 0:  
        print("Fizz", end="")  
    if num % 5 == 0:  
        print("Buzz", end="")  
    if (not num % 3 == 0) and (not num % 5 == 0):  
        print(num, end="")  
    print("")
```

Tricky because Python automatically puts newline after print command.

Discussion 5A: Print the Pattern

Read an integer width and print a pattern.

For example, if the input is 4, print

```
*  
**  
***  
****  
***  
**  
*
```

Start with nested for loops.

How many total numbers do we print?

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$$\sum_{i=1}^{i \leq n} i = \frac{n(n+1)}{2} \quad (1)$$

How many total numbers do we print?

$$\sum_{i=1}^{i \leq n} i = \frac{n(n+1)}{2} \quad (1)$$

Sketch of proof (attributed to Gauss): if $n = 100$ can pair $(1, 100)$, $(2, 99)$, etc. every pair, always adds to $n + 1$. There are $\frac{n}{2}$ of them.

How many total numbers do we print?

$$\sum_{i=1}^{i \leq n} i = \frac{n(n+1)}{2} \quad (1)$$

Sketch of proof (attributed to Gauss): if $n = 100$ can pair $(1, 100)$, $(2, 99)$, etc. every pair, always adds to $n + 1$. There are $\frac{n}{2}$ of them.

But we're summing 1 to n and $n - 1$ to 1, so we have a total of:

$$\sum_{i=1}^{i \leq n} + \sum_{i=1}^{i < n} = \frac{n(n+1)}{2} + \frac{(n-1)n}{2} \quad (2)$$

$$= \frac{n^2}{2} + \frac{n}{2} + \frac{n^2}{2} - \frac{n}{2} \quad (3)$$

$$= n^2 \quad (4)$$

Discussion 5A: Nested-Loop Version

```
width = int(input("Please enter pattern width: "))

for row in range(1, width + 1):
    for ii in range(row):
        print("*", end="")
    print()

for row in range(width - 1, 0, -1):
    for ii in range(row):
        print("*", end="")
    print()
```

Discussion 5B: Fewer Loops

Can you produce the same left-aligned pattern with only **two** for loops total?

Discussion 5C: Two-Loop Version

```
width = int(input("Please enter pattern width: "))

for row in range(1, width * 2):
    if row < width:
        star_count = row
    else:
        star_count = width * 2 - row
    for ii in range(star_count):
        print("*", end="")
    print()
```

Discussion 5C: Bonus

Bonus: can you produce the pattern with only **one** for loop?

Discussion 5C: Bonus One-Loop Version

```
width = int(input("Please enter pattern width: "))

for row in range(1, width * 2):
    if row < width:
        star_count = row
    else:
        star_count = width * 2 - row
    print("*" * star_count)
```

Discussion 5D: Bonus Centered Version

Bonus: what if we want a centered pattern instead?

```
  *  
 * *  
* * *  
* * * *  
 * * *  
  * *  
   *
```

Discussion 5D: Centered Solution

```
width = int(input("Please enter pattern width: "))

for stars in range(1, width + 1):
    print(" " * (width - stars) + " ".join("*" * stars))

for stars in range(width - 1, 0, -1):
    print(" " * (width - stars) + " ".join("*" * stars))
```

Takeaways

- Loops become easier to read when iterator names describe the role they play.
- Sentinel loops separate the stopping condition from the body of the computation.
- `continue` skips the rest of the current iteration, while `break` exits the loop entirely.
- Python has several equivalent styles: direct loops, membership tests, and comprehensions.
 - ▶ We're focusing on loops now because that's how underlying low-level code works
 - ▶ You should be able to read code with more advanced syntax
 - ▶ And know how expensive those operations are